CCC-C-425
SEPTEMBER 5, 1958
SUPERSEDING
Fed. Spec. CCC–C–411
June 28, 1932
(See 6.3)

FEDERAL SPECIFICATION

CLOTH, COTTON, BIRDSEYE, AND GAUZE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification governs the requirements of two types of cotton cloth for use in the fabrication of diapers.

1.2 Classification.

1.2.1 *Types.*—The cloth shall be of the following types as specified (see 6.2):

Type I.—Birdseye—single layer.

Type II.—Gauze—double layer.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

CCC–T–191—Textile Test Methods.

PPP–P–51 — Packaging, Packing and Marking of Textile Fabrics (Woolens, Worsteds, Cottons, Silks and Synthetics).

Federal Standards:

Fed. Std. No. 4—Glossary of Fabric Imperfections.

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes and Appendix Thereto.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample approval.—Unless otherwise specified, before production is

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commenced the contractor shall submit for approval to the contracting officer a 2-yard sample of type I cloth or a 4-panel length of type II cloth which he proposes to furnish (see 6.2).

3.2 Material.

3.2.1 *Cotton.*—The cotton shall be of a suitable length of staple and grade to meet the requirements of this specification.

3.2.2 *Yarn.*—The cotton used in the manufacture of the yarn shall be thoroughly cleaned, well carded, drawn and evenly spun into yarns to meet the requirements of this specification.

3.2.3 *Color.*—The color shall be bleached white to match an approved shade standard (see 6.4).

3.2.3.1 *Matching.*—The color shall match the approved standard shade under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard shade under incandescent lamplight at 2800° Kelvin.

3.2.4 *Weave.*

3.2.4.1 *Type I.*—The type I cloth shall be a single-layer fabric in a birdseye weave.

3.2.4.2 *Type II.*—The type II cloth shall be a double-layer gauze cloth woven in a plain, birdseye or interlocking-type weave. Pinking bars formed by joining the two layers together into a single layer shall be woven at regular intervals (see 3.2.4.2.1).

3.2.4.2.1 *Pinking bar.*—A fillingwise bar (pinking bar) 1½ ± ½ inches wide shall be woven into the cloth at regular intervals so as to form panels of the double layer (2-ply cloth). The 1½-inch pinking bars shall be made by weaving the warps of the 2-ply cloth to form a single-layer fabric.

3.2.4.2.2 *Panels.*—The panels of 2-ply cloth formed by the pinking bars shall measure 19 ± ½ inches in length from inside of pinking bar to inside of pinking bar.

3.2.4.2.3 *Folding stripes.*—The contractor may at his option, insert folding stripes. If folding stripes are used, one stripe consisting of four ends weaving as one in a group of 3 or 4 shall be positioned 13½ ± ½ inches from one selvage. Another stripe shall be positioned 8 ± ½ inches from the opposite selvage. The folding stripes shall become part of the single layer pinking bars at point of intersection.

3.2.5 *Physical requirements.*—The finished cloths shall conform to the requirements shown in table I or II as applicable.

3.2.6 *Shrinkage.*—The sum of the warp and filling shrinkage for type I or type II cloth shall not exceed 20 percent when tested as specified in table VI.

3.2.7 *Absorbency.*—The type I and type II cloths shall show a minimum absorption of 80 percent when tested as specified in table VI.

3.2.8 *Finish.*—The type I and type II fabrics shall be thoroughly cleaned, bleached, and shall contain not more than 2.0 percent total nonfibrous materials when tested as specified in table VI.

3.2.8.1 *Bleaching.*—The type I and type II cloths shall be given a full chemical bleach. The use of optical bleaches and dyes is allowed subject to approval by the contracting officer.

3.2.8.1.1 *Certificate of compliance.* — The contractor shall furnish a certificate of compliance as follows:

> It is certified that this cloth has been fully bleached and that the bleaches or dyes have been approved by the contracting officer.
> Contractors name _____.

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TABLE I.—*Physical requirements—type I, birdseye*

Weight, minimum (ounces/sq. yd.)	Width (inches)	Yarns per inch, minimum		Breaking strength, minimum (pounds)	
		Warp	Filling	Warp	Filling
4.0	27 ± ½	60	48	40	70

TABLE II.—*Physical requirements—type II, gauze*

Weight [1], minimum (ounces/sq. yd.)	Width (inches)	Yarns per inch, minimum[2]	Breaking strength [1], minimum (pounds)	
		Sum of Warp and Filling	Warp	Filling
3.35	40 ± ½	74	45	25

[1] Utilizing two plys of panel.
[2] Determined on single ply of panel, but both single panels shall conform to this requirement.

3.2.9 *Length.*—Unless otherwise specified, the cloth shall be furnished in one continuous piece of not less than 40 yards (see 6.2).

3.2.9.1 *Pieces.*—Pieces of type II cloth shall begin and end with a pinked edge located in the middle of the pinking bar.

3.2.10 *Identification.*—Each roll of cloth shall be marked for identification in accordance with Federal Specification PPP–P–51, except that manufacturers identification number and date of manufacture is not required.

3.2.11 *Workmanship.*—The finished cloth shall be clean, evenly woven and shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed that established by the acceptable quality level.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Nonmilitary.** — Nonmilitary agencies may use sampling, inspection and test procedures outlined under 4.2 when desirable. When procedures outlined under 4.2 are not deemed suitable, the procedures specified under nonmilitary paragraphs shall be followed.

4.1.1 *Sampling.*

4.1.1.1 *Inspection lot.* — Unless otherwise specified in the invitation for bids, all cloth of one type submitted at one time shall be considered a lot for purposes of inspection and test.

4.1.1.2 *Sampling for inspection.*—A random sample of cloth shall be taken from each lot in accordance with table III.

4.1.1.3 *Sampling for tests.*—Random samples for tests shall be taken from each inspection lot in accordance with table IV.

4.1.2 *Preproduction sample inspection.*—When a preproduction sample is required, it shall be inspected and tested as specified in 4.2.2.1.

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TABLE III.—*Sampling for inspection*

Number of yards in lot	Number of rolls or bolts in the inspection sample	Number of yards in the inspection sample [1]	Acceptance Number total (defective)	Rejection Number total (defective)	Acceptance Number major (defective)	Rejection Number major (defective)
300 and under	3	35	10	11	3	4
301 to 500	3	50	13	14	4	5
501 to 800	3	75	19	20	6	7
801 to 1,300	3	110	26	27	8	9
1,301 to 3,200	5	150	34	35	11	12
3,201 to 8,000	7	225	48	49	17	18
8,001 to 22,000	10	300	63	64	20	21
22,001 to 110,000	15	450	89	90	29	30
110,001 to 550,000	25	750	89	90	45	46
550,001 and over	50	1,500	89	90	81	82

[1] An approximately equal number of yards shall be examined from each piece in the sample.

4.1.3 *Yard-by-yard examination.*—Each of the samples selected in accordance with table III, shall be visually inspected and defects classified in accordance with 4.2.3.1.1. The maximum number of defects acceptable shall be as specified in table III.

4.1.4 *Over-all inspection and inspection for length.*—Over-all inspection and inspection for length shall be as specified in 4.2.3.1.2 and 4.2.3.1.3, respectively. Sample-size and the maximum number of defects acceptable shall be as specified in table V.

4.1.5 *Tests.*—Samples selected for test in accordance with table IV shall be tested in accordance with the applicable methods of Federal Specification CCC–T–191 as listed in table VI of this specification. The physical values specified in section 3 of this specification apply to the average of the determinations made on a unit of product for test purposes as specified in the applicable test methods. The unit of product shall be three continuous yards full width of finished cloth.

TABLE IV.—*Sampling for tests*

Number of rolls or bolts in lot	Number of rolls or bolts in sample[1]	Acceptance No. (defective)	Rejection No. (defective)
8 and under	2	0	1
9 to 15	3	0	1
16 to 40	4	0	1
41 and over	5	0	1

[1] One unit of product (test sample) at least 3 yards long and the full width of the cloth shall be taken from each sample roll.

4.1.6 *Compliance criteria.*

4.1.6.1 *Inspection.*—Any piece in the sample for inspection failing to conform to the requirements for which the inspection was made shall be considered defective. If the number of defective pieces in the sample exceeds the acceptance number for that sample, the entire lot from which the sample was taken shall be considered to have failed to meet the requirements of this specification.

4.1.6.2 *Tests.*—Any piece in the sample for tests failing to conform to the requirements for which tests were made shall be considered defective. If the number of defective pieces in the sample exceeds the acceptance number for that sample, the entire lot from which the sample was taken shall be considered to have failed to meet the requirements of this specification.

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4.3 Testing of the end item.—The methods of testing specified in Federal Specification CCC–T–191, wherever applicable, and as listed in table VI shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a unit of product for test purposes as specified in the applicable test methods. The unit of product shall be three continuous yards, full width, of the finished cloth. For each characteristic, the acceptable quality level shall be 6.5 test failures per 100 units (yards) and the inspection level shall be level L–1, table VII of the Appendix to Military Standard MIL–STD–105. The lot size shall be expressed in terms of one yard each.

TABLE VI.—*Tests*

Characteristic	Requirement reference	Test method
Identification of cotton	3.2.1	1200
Weave	3.2.4	Visual
Weight, oz. sq. yd.[1]	Table I	5041
Yarns per inch[2]	Table I	5050
Breaking strength[3]	Table I	5100
Shrinkage in laundering[4]	3.2.6	5550
Absorbency	3.2.7	5502
Nonfibrous materials[5]	3.2.8	2611

[1] For type II—determination to be made on the two layers of body cloth, and should not include folding stripes if present.

[2] For type II—determination to be made on single layer of panel, but both layers must meet requirement, specimen should not include folding stripes if present.

[3] For type II—determination to be made on double layer body cloth (not single ply). All tests must be clear of and must not include the pinking bars of folding stripes if present.

[4] The test shall be conducted in accordance with Method 5550 except as follows: Three specimens shall be prepared for test. The specimens of type I cloth shall be cut 26 inches in length, the full width of the cloth. The specimens of type II cloth shall be cut the width of the cloth through the middle of the respective pinking bars leaving a pinked edge. The samples shall be dried prior to pressing.

[5] The total nonfibrous material shall be determined by Method 2611 utilizing the following formula:

$$\text{Nonfibrous materials content, percent} = \frac{O\text{–}S \times 100}{O}$$

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *Levels A, B, and C.*—Cloth of one type only shall be packaged in accordance with the applicable requirements for class 3 fabrics of Federal Specification PPP–P–51.

5.2 Packing.

5.2.1 *Levels A, B, and C.*—Cloth of one type only shall be packed in accordance with the applicable requirements of Federal Specification PPP–P–51.

5.3 Marking.

5.3.1 *Civil agencies.*—In addition to any special marking specified in the contract or order, each shipping container shall be marked in accordance with Federal Standard No. 123.

5.3.2 *Military agencies.*—Shipments shall be marked in accordance with the applicable requirements of Federal Specification PPP–P–51.

5.4 Federal Standard No. 102 shall be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment (civil agencies only).

6. NOTES

6.1 Intended use.—The cloths governed by the requirements of this specification are intended for use in the fabrication of infants diapers.

6.2 Ordering data.—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type required.
c. When preproduction sample is not required (see 3.1).
d. Length when other than specified (see 3.2.9).

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e. Both the level of packaging and the level of packing required (see 5.1 and 5.2).

6.2.1 In the preparation of contracts or orders, it must be noted that various options, choices and alternatives, as indicated in Federal Specification PPP–P–51, may be exercised in the preparation for delivery of cloth.

6.3 Supersession data.—This specification supersedes Federal Specification CCC–C–411 for Cloth, Birdseye (Diaper), in Bolts and includes new requirements for the gauze diaper fabric.

6.4 Standard sample.—For access to standard sample, address the procuring office issuing the invitation for bids.

6.5 Transportation description.—The transportation description applicable to this item is:

Cotton piece goods, in the original piece, not otherwise indexed by name. (Diaper cloth.)

Carload minimum weight, any quantity.

Motor volume minimum weight, any quantity.

Notice. — When Government drawings, specification, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
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4.2 Military.

4.2.1 *General.*—The quality assurance provisions contained in this section are applicable to inspection by and for the Government. Quality assurance provisions for components or materials shall be in accordance with this specification and subsidiary specifications referenced herein to the extent applicable, except that this specification shall govern in the event of conflict.

4.2.2 *Preacceptance inspection.*

4.2.2.1 *Preproduction sample inspection.*—The preproduction sample submitted in accordance with 3.1 shall be visually inspected for appearance, color, and finish. The sample

Glossary No. Fed. Std. No. 4	Defects	Classification	
		Major	Minor
90–91–92	Hole, cut or tear in body of cloth	X	
78	Smash, any	X	
11–33	Broken or missing yarns:		
	Three or more contiguous regardless of length	X	
	Two contiguous, or single 9 inch or more missing		X
9	Fine filling bar or light place, clearly noticeable	X	
9–12–35	Open place, crack or hitchback-opening at widest place four or more yarn diameters or sequence of openings resulting in weak spot	X	
13–34	Floats or skips:		
	Multiple, ¾ inch or more in combined warp and filling direction	X	
	Multiple, less than ¾ inch in combined warp and filling directions, or single 2 inches or more in length		X
3–5–27	Slough-off, jerked in filling slubs, slugs or gouts—three or more times the thickness of normal yarns		X
6–7–28–29	Kinks or knots, noticeable accumulation		X
59–61	Crease or wrinkle—hard, embedded		X
	Narrow width		X
48–53–54	Selvage curled, rolled, folded, stringy or loopy		X
55–56	Selvage cut, torn or broken	X	
51	Selvage scalloped—deepest indentation more than ½ inch from straight edge of cloth		X
74–75–76	Spot, stain or streak		X
83	Trash—motes, seed coat fragment or leaf—any noticeable accumulation		X

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shall be tested for physical and chemical properties in accordance with the methods shown in table V when applicable.

4.2.3 *Inspection for acceptance.*—Inspection shall be in accordance with the provisions of Military Standard MIL–STD–105 and Appendix except where otherwise indicated.

4.2.3.1 *Inspection of the end item.*—Defects found during this examination shall be classified in accordance with the list shown in 4.2.3.1.1, 4.2.3.1.2 and 4.2.3.1.3.

4.2.3.1.1 *Yard by yard examination.*—The required yardage shall be examined and visual defects classified as listed. All defects found shall be counted regardless of their proximity to each other except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The unit of product for this examination shall be one linear yard. The acceptable quality level (AQL) for this examination shall be 4.0 major defects and 15 total defects (major and minors combined) per 100 units (yards). The lot size shall be based upon level II, table III of Military Standard MIL–STD–105. The lot size shall be expressed in units of one yard each. The number of pieces from which the sample is to be selected shall be in accordance with table V. An approximately equal number of yards shall be examined from each piece in the sample.

TABLE V.

Lot size in yards	Sample size in pieces	Maximum number of defects acceptable in sample
Up to 1300[1]	3	0
1301 to 3200	5	0
3201 to 8000	7	0
8001 to 22,000	10	1
22,001 to 110,000	15	2
110,001 to 550,000	25	2
550,001 and over	50	3

[1] If a lot contains fewer than three pieces, each piece in the lot shall be examined.

4.2.3.1.2 *Over-all examination.*—Each defect listed shall be counted not more than once in each piece examined. The unit of product for this examination shall be one piece. The sample size and acceptance number shall be as shown in table V.

Defects

Selvage curled, folded, rolled, or slack continuously or intermittently throughout the piece.

Over-all uncleanness, objectionable odor.

Width, less than specified.

Uneven weaving throughout the piece.

Not bleached white.

Type II only:

Length of panels not within tolerances.

Width of pinking bar not within tolerances.

Folding stripes — if present — not in specified locations.

4.2.3.1.3 *Examination for length.*

4.2.3.1.3.1 *Individual pieces.* — The pieces shall be examined for gross length. Any gross length found to be less than the specified minimum length, or any gross length found to be more than two yards below the gross length marked on the piece ticket shall be considered as a defect with respect to length. The unit of product for this examination shall be one piece. The number of pieces selected as sample for this examination and the number of defects acceptable shall be in accordance with table V.

4.2.3.1.3.2 *Total yardage in sample.*—The lot shall be unacceptable if the total of the actual gross length of pieces in the sample is less than the total of the gross lengths marked on piece tickets. The pieces examined shall be those selected for examination of individual pieces.

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